Continuity, Piecewise Continuity, and Discontinuities
An Introduction from the Perspective of Real Analysis

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1 Introduction

One of the central goals of real analysis is to understand how functions behave. Some functions vary smoothly, while others exhibit abrupt changes, oscillations, or singularities. The concept of continuity provides a precise mathematical language for distinguishing these behaviors.

Although continuity is introduced early in calculus, its true significance becomes apparent only in analysis. Many of the fundamental theorems of calculus are, at their core, statements about continuous functions. Conversely, studying the different ways in which continuity can fail reveals why certain hypotheses are necessary.

The purpose of this chapter is to develop both an intuitive and formal understanding of

- continuous functions,
- discontinuities,
- piecewise continuous functions,
- and their role in integration and differential equations.

Throughout, we work with functions

$$f : D \to \mathbb{R},$$

where the domain $D \subseteq \mathbb{R}$.

2 Limits: The Foundation of Continuity

Continuity is defined in terms of limits. Consequently, the notion of a limit must come first.

Definition 2.1 (Limit)**.** Let a be an accumulation point of the domain of f.

We say

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Notice that the value $f(a)$ does not appear in the definition.

A limit depends only on the behavior of the function arbitrarily close to the point.

This distinction is one of the most important ideas in elementary analysis.

3 Continuous Functions

The notion of continuity is obtained by requiring that the function value agree with its limiting value.

Definition 3.1 (Continuity at a Point)**.** A function f is continuous at a if

$$\lim_{x \to a} f(x) = f(a).$$

Intuitively, nearby inputs produce nearby outputs.

The function behaves predictably under arbitrarily small perturbations of the input.

Definition 3.2 (Continuous on an Interval)**.** A function is continuous on an interval if it is continuous at every point of that interval.

4 Equivalent Characterizations

Real analysis offers several equivalent ways to express continuity.

Theorem 4.1. *For a function $f : \mathbb{R} \to \mathbb{R}$, the following are equivalent.*

1. *f is continuous at a.*
2. *Whenever $x_n \to a$,*

$$f(x_n) \to f(a).$$

3. *The inverse image of every open set is open (topological definition).*

The second characterization is often easier to use in proofs, while the third extends naturally to general metric and topological spaces.

5 How Continuity Can Fail

Discontinuities arise whenever

$$\lim_{x \to a} f(x) \neq f(a)$$

or the limit does not exist.

However, the failure may occur in several fundamentally different ways.

5.1 Removable Discontinuities

Example 5.1. Consider

$$f(x) = \frac{x^2 - 1}{x - 1}, \qquad x \neq 1.$$

Since

$$\frac{x^2 - 1}{x - 1} = x + 1,$$

we obtain

$$\lim_{x \to 1} f(x) = 2.$$

The limit exists, but the function is not defined at $x = 1$.

By defining

$$f(1) = 2,$$

the discontinuity disappears.

A removable discontinuity represents missing information rather than genuinely irregular behavior.

5.2 Jump Discontinuities

A jump discontinuity occurs when the one-sided limits both exist but differ.

Example 5.2.

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} f(x) = 0, \qquad \lim_{x \to 0^+} f(x) = 1.$$

Since these limits are unequal, the ordinary limit does not exist.

5.3 Infinite Discontinuities

Infinite discontinuities occur when the function grows without bound.

The simplest example is

$$f(x) = \frac{1}{x}.$$

Near the origin,

$$|f(x)| \to \infty.$$

5.4 Oscillatory Discontinuities

Perhaps the most important counterexample in introductory analysis is

$$f(x) = \sin\left(\frac{1}{x}\right).$$

As $x \to 0$, the oscillations become infinitely frequent.

No limiting value exists.

Unlike a jump discontinuity, there is no preferred left or right value.

6 One-Sided Limits

Many practical functions possess jumps.

To describe them, we introduce one-sided limits.

Definition 6.1. The left-hand limit is

$$\lim_{x \to a^-} f(x),$$

while the right-hand limit is

$$\lim_{x \to a^+} f(x).$$

Theorem 6.2. *The ordinary limit*

$$\lim_{x \to a} f(x)$$

exists if and only if

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x).$$

7 Piecewise Continuous Functions

Many functions encountered in applications are continuous except at isolated jump discontinuities.

Definition 7.1. A function is piecewise continuous on a closed interval $[a, b]$ if

1. it has only finitely many discontinuities;
2. on every open subinterval between consecutive discontinuities it is continuous;
3. every discontinuity has finite left and right limits.

For functions defined on

$$[0, \infty),$$

the definition requires every finite interval

$$[0, b]$$

to satisfy the above conditions.

Consequently, infinitely many discontinuities are allowed over the entire half-line provided they do not accumulate inside any finite interval.

8 Examples

Example 8.1. Every polynomial is continuous.

Example 8.2. Every rational function is continuous wherever its denominator is nonzero.

Example 8.3. The floor function

$$\lfloor x \rfloor$$

is piecewise continuous.

It has jump discontinuities at every integer.

Example 8.4. The sign function

$$\mathrm{sgn}(x)$$

is piecewise continuous.

Example 8.5. The function

$$\sin\left(\frac{1}{x}\right)$$

is not piecewise continuous on any interval containing the origin.

9 Important Theorems About Continuous Functions

The following theorems form the backbone of elementary analysis.

Proofs are omitted.

Theorem 9.1 (Algebra of Continuous Functions)**.** *Sums, differences, products, and scalar multiples of continuous functions are continuous.*

Whenever defined, quotients of continuous functions are continuous.

Theorem 9.2 (Composition)**.** *The composition of continuous functions is continuous.*

Theorem 9.3 (Extreme Value Theorem)**.** *Every continuous function on a compact interval*

$$[a, b]$$

attains both a maximum and a minimum.

Theorem 9.4 (Intermediate Value Theorem)**.** *If*

$$f(a) < c < f(b),$$

then there exists

$$x \in (a, b)$$

such that

$$f(x) = c.$$

Theorem 9.5 (Uniform Continuity)**.** *Every continuous function on a compact interval is uniformly continuous.*

10 Continuity and Integration

One reason continuity is so important is its relationship with integration.

Theorem 10.1. *Every continuous function on a closed bounded interval is Riemann integrable.*

The converse is false.

Many discontinuous functions are nevertheless Riemann integrable.

For example, every piecewise continuous function on a closed interval is Riemann integrable.

Thus jump discontinuities do not destroy integrability.

11 Continuity and Differentiability

Differentiability is a stronger condition than continuity.

Theorem 11.1. *If f is differentiable at a, then f is continuous at a.*

The converse does not hold.

The classical example is

$$f(x) = |x|,$$

which is continuous everywhere but not differentiable at the origin.

12 A Hierarchy of Regularity

Functions encountered in analysis can be arranged according to increasing regularity.

$$\begin{aligned} &\text{arbitrary functions} \\ &\supset \text{Riemann integrable} \\ &\supset \text{piecewise continuous} \\ &\supset \text{continuous} \\ &\supset C^1 \\ &\supset C^2 \\ &\supset \cdots \\ &\supset C^\infty \\ &\supset \text{analytic.} \end{aligned}$$

Each level imposes additional structure.

As assumptions become stronger, more powerful theorems become available.

13 Why Piecewise Continuity Appears So Often

Many physical systems exhibit abrupt but isolated changes.

Examples include

- electrical switches,
- relay circuits,
- digital signals,
- square waves,
- pulse trains,
- piecewise constant forcing terms in differential equations.

These systems are not continuous, yet they remain sufficiently regular for the machinery of calculus to apply.

Consequently, piecewise continuity is a standard hypothesis in the theory of Laplace transforms, Fourier series, and differential equations.

14 Summary

The concept of continuity formalizes the intuitive idea that small changes in the input produce small changes in the output.

Discontinuities arise in several qualitatively different ways, including removable, jump, infinite, and oscillatory discontinuities.

Among discontinuous functions, piecewise continuous functions occupy an especially important position. They preserve many of the useful analytical properties of continuous functions while allowing isolated jumps that naturally occur in applications.

Understanding continuity is therefore not merely an exercise in manipulating limits. It is the first step toward understanding why the major theorems of analysis are true and precisely which hypotheses they require.